

Michael Sipes

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Education

University of Southern California

M.S. in Computer Science

2025 - 2026

B.S. in Computer Science

2021 - 2025

Relevant Coursework: Machine Learning, Natural Language Processing, Analysis of Algorithms, Data Structures, Computer Systems, Operating Systems, Internetworking, Web Technologies, Software Engineering, Database Systems

Languages: C++, C, Python, Java, JavaScript, TypeScript, HTML, CSS, SQL, Bash, Markdown, LaTeX, Assembly

Frameworks / Technologies: React, Next.js, OpenAI API, Vercel AI SDK, Supabase, Vercel, Zustand, Tailwind CSS, PyTorch, Docker, Git

Experience

RingCentral

AI Software Engineering Intern

June 2024 - August 2024

- Built a retrieval-augmented generation system to detect and consolidate duplicate content across a knowledge base of 2,000+ articles.
- Conducted interviews with article writers, understood their workflow challenges, and defined technical requirements for the solution.
- Implemented the solution by generating semantic embeddings with OpenAI and storing them in Supabase (pgvector extension) to enable similarity search.

Active Owners Fund

Financial Analyst Intern

May 2023 - August 2023

- Performed credit analysis through sophisticated financial modeling to support debt fund analysts.
- Developed machine learning models using cross-validation, linear and logistic regression, regularization, clustering, random forests, gradient descent, and neural networks.
- Learned PyTorch for deep learning, covering neural network classification, computer vision, custom datasets, transfer learning, and model deployment.

Academic Projects

DrawTunes

Live Demo | GitHub

- Built AI-powered music discovery platform where users draw artwork to receive personalized song recommendations using React, TypeScript, and Supabase edge functions.
- Refactored state management migration from React Context to Zustand, reducing unnecessary re-renders and optimizing performance across interconnected components.
- Implemented modular serverless architecture processing image uploads through OpenAI Vision API and iTunes API integration for real-time music recommendation delivery.

Layers

Live Demo | GitHub

- Built AI-powered skiing outfit recommendation app that learns from user feedback to provide personalized clothing recommendations based on personal comfort data and weather conditions.
- Implemented dual interface architecture where both traditional UI and AI chat perform identical database operations, ensuring feature parity and consistent user experience across interaction modes.
- Integrated real-time weather data and location services to automatically adjust recommendations based on current conditions at user's ski resort.

Mjsipes Photography

Live Demo | GitHub

- Built personal photography portfolio using Next.js and Supabase Storage to match Squarespace display quality while adding AI-generated artistic reinterpretations of 300 personal photographs.
- Implemented dynamic page routing and metadata database architecture designed for experimental workflows, tracking relationships between original photos and AI style variations.
- Implemented automated image processing pipeline using Node.js scripts and OpenAI Images API to generate 1,800 style-transfer variations across 6 artistic styles.

Honors

Sydney Harmon Academy for Polymathic Study Fellow 2025, Effective Altruism Fellow 2022